Lab 5 · Pathogenwatch and Microreact: Many Genomes at Once

Day 2 · 11:30–12:30 · 60 min · Pairs Objective: LO5 — Use free web platforms to type, screen and visualise a collection of genomes

> **Scenario:** The IPC nurse at *District Hospital A (fictional)* is worried about a rise in MRSA SSIs on Surgical Ward B. The lab stored the isolates and sent them for Illumina sequencing, and your colleague assembled them in Galaxy. You now have **genomes from all the MRSA isolates from the past months** and want a quick overview. *The genomes are real public data. The hospital, wards and dates are fictional.*

Files from the instructor, in the `labs/data/` folder of the course GitHub (https://github.com/aryakaul/ughe-ssi-genomics-course):

- `assemblies/` — one FASTA per isolate
- `koser_tree.nwk` — phylogenetic tree (built in Galaxy with Snippy → IQ-TREE)
- `lab6_metadata.csv` — patient, ward, procedure and date information

> **Download everything at once:** on https://github.com/aryakaul/ughe-ssi-genomics-course click the green **Code** button → **Download ZIP**, then unzip it. The files are in `labs/data/`. *Alternative:* the shared Galaxy history https://usegalaxy.eu/u/aryakaul/h/ughe-ssi-course-precomputed (click a dataset → **download** icon 💾).

Part A — Pathogenwatch: automatic typing and AMR (30 min)

Pathogenwatch (https://pathogen.watch) is a free platform run by the Centre for Genomic Pathogen Surveillance. It identifies species, sequence type and AMR, and places genomes in global context, **without any coding**.

A1. Upload

1. Log in to **https://pathogen.watch** (account created before the course)
2. Click **Upload** and drag in the FASTA assemblies
3. Wait while each genome is analysed (a progress bar appears; usually a few minutes)

A2. Explore the genome list

For each genome, Pathogenwatch reports:

- **Organism**: does it agree with the lab?
- **ST** (MLST)
- **AMR**: predicted resistance by drug, and the genes responsible
- Assembly **QC metrics** (length, contigs, N50)

Fill in a table for **at least 6** genomes:

Genome	Species	ST	Predicted resistances	QC pass?

A3. Make a collection and view the tree

1. Select all your uploaded genomes → **Create collection** → name it `UGHE Lab 5`
2. Open the collection and explore the **tree** view
3. Explore the **AMR** view, which shows resistance profiles side by side

✍️ **Check yourself A:**

1. How many different STs are in the collection?
2. Do all isolates have the **same** AMR profile? What does that suggest, and what does it **not** prove?
3. In the tree, can you see a group of isolates on very **short** branches?

> 💡 Pathogenwatch also lets you compare your genomes with **public genomes from around the world**. Try **"Find similar"** or the population view, if the instructor enables it.

Part B — Microreact: tree + timeline + metadata (25 min)

Microreact (https://microreact.org) combines a **tree**, a **timeline**, a **map** and a **table** in one interactive view. It's ideal for presenting results to an IPC committee.

B1. Create a project

1. Go to **https://microreact.org** → **Upload** (you can sign in with Google or another account if you want to save the project; viewing is possible without it)
2. Drag in **both** files: `lab6_metadata.csv` and `koser_tree.nwk`
3. Microreact asks which column links the table to the tree tips → choose **`id`**
4. If asked about dates, confirm that **`year`**, **`month`**, **`day`** form the date
5. Click **Continue / Create**

B2. Explore

- **Colour** the tree by **`ward`**: click the colour-by control and choose `ward`
- Then colour by **`theatre`**, and then by **`hospital`**
- Open the **Timeline** panel: when did each infection happen?
- **Click a tip** in the tree: the matching row is highlighted in the table and timeline
- **Lasso-select** a group of closely related tips: where and when do they fall?

✍️ **Check yourself B:**

1. Is there a group of isolates that are closely related **and** close in time **and** in the same place?
2. Is there any isolate from the **same ward at the same time** that is **not** in that group?
3. Is there any isolate in the group that came from a **different ward**? What could link it?

B3. Save and share

- Click **Save** (if signed in) and copy the project link. You'll use it in **Lab 6**.

Summary

Tool	Gives you	Input
Pathogenwatch	Species, ST, AMR, QC, clustering, global context	FASTA assemblies
Microreact	Visual story: tree + time + place + metadata	Tree (Newick) + table (CSV)

Both are free and used by national public-health labs around the world. **You can use them on Monday with your own data.**

Pathogenwatch: Centre for Genomic Pathogen Surveillance. Microreact: Argimón S, et al. Microb Genom 2016.